

Amit Kumar Mohanty

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PROFESSIONAL SUMMARY

AI Engineer with about 1.5 years of experience, currently pursuing an M.Tech in AI and ML at BITS Pilani. As a Research Associate, AI at Keywords Studios, I work across LLMs, Agentic AI, and Generative AI to build intelligent systems that feel genuinely useful, and was named "Rookie of the Year 2025" for surpassing every performance KPI. With earlier internships at Infosys Springboard and QL2 Software, plus a B.Tech in Computer Science (AI and ML) from K.R. Mangalam University, I love turning promising ideas into reliable, real world products.

EDUCATION

Birla Institute of Technology and Science, Pilani (BITS Pilani)

July 2026 – July 2028

M.Tech in Artificial Intelligence and Machine Learning (Currently Pursuing)

Relevant Coursework: NLP, Deep Learning, Generative AI, LLMs, Deep Neural Networks, Deep Reinforcement Learning

K.R. Mangalam University

July 2021 – July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana

May 2025 – Present

- Awarded "**Rookie of the Year 2025**" for surpassing **all KPIs** through advanced model training, data-pipeline optimization, and cross-functional collaboration.
- Trained and optimized **enterprise-grade Agentic AI** models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving **task-completion reliability**.
- Contributed to a cutting-edge **web-automation AI agent** that visually interprets on-screen content and, from a natural-language instruction, autonomously plans and executes browser actions such as clicking, typing, and navigating to complete complex, multi-step tasks.
- Trained, evaluated, and quality-controlled the agent with a **global team**, curating high-quality datasets and running structured test reviews to deliver client-ready iterations with **improved action accuracy and reliability**.
- Built custom data pipelines and delivered **85%** of self-generated training datasets directly to production as the **first member of a global AI team**, accelerating model readiness.

Artificial Intelligence Intern, Infosys – Remote

November 2024 – February 2025

- Engineered an **AI call-center system** with LangChain and LangSmith, boosting response accuracy by **35%** and elevating customer-AI interaction quality.
- Developed and deployed LLM features that reduced system latency by **25%**, streamlining the AI deployment pipeline.
- Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing **hallucinations by 20%** and making customer conversations feel more natural and trustworthy.
- Collaborated with **12 engineers** to integrate end-to-end AI workflows and was selected to present at the **Best Project by Indian Interns** initiative.

PROJECTS

Dynamic Screen Companion

June 2025 – Present

- Built a **real-time multi-modal AI assistant** integrating the **Gemini API** to analyze live screen activity and deliver continuous voice feedback and interactive support.
- Engineered a high-performance FastAPI model-serving backend handling concurrent inference streams, reducing manual task time by **40%**.
- Designed the end-to-end multi-modal architecture from scratch and optimized OCR pipelines to maintain **95%+ extraction accuracy** across varied layouts.

Diabetic Optiscan

January 2025 – June 2025

- Built an AI tool that scans retinal (eye) images to detect early signs of diabetic retinopathy, and co-authored a **peer-reviewed research paper** explaining how the system was designed and the accuracy improvements it achieved.
- Led integration of a **Vision Transformer (ViT-B/16)** on the APTOS dataset, optimizing self-attention to capture complex spatial patterns in retinal images.
- Designed a custom wavelet modification technique that boosted **detection precision by 50%** over legacy methods.

Currency Detection App for Visually Impaired

July 2024 – December 2024

- Trained an image-classification model on **200,000+ banknote images** achieving **96% accuracy**, deployed in a mobile app with voice feedback to help prevent fraud.
- Conducted **user interviews with visually impaired individuals** to refine real-world accessibility features.

SKILLS

Languages & Frameworks: Python, C++, SQL, FastAPI, TensorFlow, PyTorch, LangChain, LangGraph, Hugging Face, OpenCV, Scikit-learn
Generative & Agentic AI: LLMs, Agentic AI, RAG, Fine-tuning (LoRA, QLoRA), Prompt Engineering, Transformers, BERT, Multi-modal AI, Gemini/OpenAI APIs, NLP, Vector Databases (FAISS/Chroma)

AI Systems & Deployment: Model Serving, API Development, Docker, CI/CD, MLflow, Model Evaluation & Optimization, Data Pipelines, Git

Data & Cloud: AWS, GCP (Vertex AI), MySQL, MongoDB, Pandas, NumPy, Power BI

CERTIFICATIONS

Ducat India – Data Science Professional Training

April 2024 – April 2025

Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.

AI/ML for Geodata Analysis – ISRO

Applied ML and deep learning to satellite and remote-sensing imagery for terrain classification and geospatial pattern analysis.